Numerical Differentiation

First-Order Derivative

We start with **Taylor expansion around x**:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots$$

Rearranging for $f'(x)$:

$$f'(x) = \frac{f(x+h) - f(x)}{h} - \frac{h}{2}f''(x) - \frac{h^2}{6}f'''(x) - \cdots$$

First-Order Derivative

Forward Difference Approximation:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

Central Difference Formula

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

Backward Difference Approximation:

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}$$

Second-Order Derivative

Use the same Taylor expansions again.

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(x) + \cdots$$
$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(x) + \cdots$$

Add them:

$$f(x+h) + f(x-h) = 2f(x) + h^2 f''(x) + \frac{h^4}{12}f^{(4)}(x) + \cdots$$

Rearranging for $f''(x)$:

$$\boxed{f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}}$$

Higher-Order Formulas for First Derivative

Use Taylor expansions

$$f(x) = f(x)$$

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \cdots$$

$$f(x+2h) = f(x) + 2hf'(x) + 2h^2f''(x) + \frac{4h^3}{3}f'''(x) + \cdots$$

$$Af(x) + Bf(x+h) + Cf(x+2h)$$

$$\begin{aligned} &= (A+B+C)\,f(x) \\ &\quad + (B+2C)h\,f'(x) \\ &\quad + \left(\frac{B}{2} + 2C\right)h^2 f''(x) \\ &\quad + \left(\frac{B}{6} + \frac{4C}{3}\right)h^3 f'''(x) + \cdots \end{aligned}$$

Higher-Order Formulas for First Derivative

Cancel unwanted terms

$$A + B + C = 0$$

$$\frac{B}{2} + 2C = 0 \quad \Rightarrow \quad B = -4C$$

$$\frac{B}{2} + 2C = 0$$

$$A - 4C + C = 0 \quad \Rightarrow \quad A = 3C$$

$$3C\, f(x) - 4C\, f(x+h) + C\, f(x+2h)$$

$$\boxed{f'(x) = \frac{-f(x+2h) + 4f(x+h) - 3f(x)}{2h} + O(h^2)}$$

$$C[-f(x+2h) + 4f(x+h) - 3f(x)]$$